SHIVAM KUMAR

Embedded Systems Engineer — ECE Undergraduate

+91-7367021106 — shivam8621116@gmail.com — LinkedIn — GitHub

PROFILE SUMMARY

Electronics and Communication Engineering undergraduate at IIIT Manipur. Experienced in hardware-software integration and firmware development using C/C++ and Python. Proven track record in deploying real-time monitoring systems on ESP32 and Arduino platforms. Seeking an embedded/firmware engineering internship to apply low-level programming expertise, system design skills, and foundational knowledge of VLSI and AI integration.

TECHNICAL SKILLS

Prog. Languages: C (Embedded), C++, Python (Intermediate), VHDL, HTML/CSS
Embedded Systems: ESP32, Arduino, Raspberry Pi, Zynq, Microcontrollers, Sensors & Actuators
VLSI & EDA Tools: Cadence, Synopsys, Digital & Analog Circuits
Tools & Software: MATLAB, GNU Octave, Keil uVision, Tinkercad, Git, VS Code
Core Concepts: Data Structures & Algorithms (C), Signal Processing , Debuging
Languages: English, Hindi, Maithili (Native)

PROJECTS

ESP32 Wearable Safety Monitoring & Location Tracking System

- **Engineered** a wearable IoT device using **ESP32, GPS, and SIM800L (GSM)** modules for real-time tracking.
- **Developed** firmware in **C** to handle asynchronous data transmission and GPS coordinate parsing.
- **Integrated** real-time monitoring protocols to ensure continuous environmental data acquisition.
- **GitHub Link:** Project Repository

Temperature & Humidity Monitoring System

- **Architected** a monitoring solution using **Arduino** and **DHT11/DHT22** sensors for climate tracking.
- **Implemented** serial communication and I2C LCD interfacing for real-time data visualization.
- **Optimized** data acquisition workflows to log environmental metrics with low latency.

EDUCATION

Indian Institute of Information Technology (IIIT), Manipur — Imphal, Manipur
Bachelor of Technology in Electronics & Communication Engineering — *2023 – 2027*

P D K J College, Andhrathari, Madhubani — Madhubani, Bihar
Higher Secondary Education (Class XII) — **82.8%**

TRAINING & WORKSHOPS

Drone Bootcamp — IIIT Manipur — *July 2025*

- **Analyzed** UAV electronics, including flight controllers, sensor fusion, and system architecture.
- **Collaborated** on circuit interfacing and sensor integration during team-based drone assembly.

Space Engineering & AI Workshops — *Oct – Dec 2025*

- **Rocket Science (India Space Academy):** Studied propulsion, staging, and introductory orbital mechanics.
- **Satellite Data Analysis:** Processed geospatial data and remote sensing interpretation using Google Earth Engine.
- **Generative AI (NxtWave):** Explored AI model workflows, implementation fundamentals, and problem-solving.